

Increment and Decrement Operations - Zero to Hero Questions List

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1. Write a C program to take an integer and print its value before and after using `++`. (*Hint: Use pre-increment and post-increment separately.*)
2. Write a program to demonstrate the difference between `a++` and `++a`. (*Hint: Print values during assignment.*)
3. Take two integers and apply increment operators to both in one expression. (*Hint: Observe evaluation order carefully.*)
4. Write a program to decrement a number using `--` and print intermediate values. (*Hint: Use both pre-decrement and post-decrement.*)
5. Swap two numbers using increment and decrement operators without using a temporary variable. (*Hint: Use arithmetic with `++` and `--` carefully.*)
6. Write a program to count from 1 to 10 using only increment operators. (*Hint: Use a `while` loop.*)
7. Write a program to count backward from 10 to 1 using decrement operators. (*Hint: Use `--` inside the loop.*)
8. Predict the output of:

```
int a = 5;
printf("%d %d", a++, ++a);
```

(*Hint: Undefined behavior concept.*)

9. Write a program to increment a variable five times using nested statements. (*Hint: Try repeated `++a`.*)
10. Create a calculator that increases the entered number by 1 and decreases it by 1. (*Hint: Use both operators together.*)
11. Write a program to check how post-increment behaves inside `printf()`. (*Hint: Print variable before and after statement.*)
12. Write a program to compare pre-increment and post-increment in assignments. (*Hint: `b = a++` vs `b = ++a`.*)
13. Find the output:

```
int x = 10;
int y = x++ + ++x;
```

(*Hint: Expression side effects.*)

14. Write a program that increments only even numbers entered by the user. (*Hint: Use `if` condition.*)
15. Write a menu-driven program for incrementing and decrementing numbers. (*Hint: Use `switch` statement.*)
16. Create a loop using only decrement operators to print odd numbers from 15 to 1. (*Hint: Decrease by 2.*)
17. Write a program to simulate a countdown timer using `--`. (*Hint: Use `sleep()` if available.*)
18. Write a program to increase array elements using increment operators. (*Hint: Traverse array with loop.*)
19. Write a program to decrement all elements of an array by 1. (*Hint: Use `arr[i]--`.*)
20. Write a program to print ASCII characters using increment operators. (*Hint: Increment character variable.*)
21. Write a program to print alphabets A–Z using `++`. (*Hint: Use character data type.*)
22. Write a program to print reverse alphabets Z–A using decrement operator. (*Hint: Use `char ch = 'Z';`.*)
23. Predict the output:

```
int a = 3;
int b = a++ + a++ + ++a;
```

(*Hint: Multiple modifications in one expression.*)

24. Write a program that increments a number until it becomes divisible by 7. (*Hint: Use loop and `%` operator.*)
25. Write a program to decrement a number until it reaches zero. (*Hint: Use `do-while` loop.*)
26. Create a simple visitor counter using increment operator. (*Hint: Increment count each iteration.*)
27. Write a program to simulate bank token numbers using post-increment. (*Hint: Token should print before increment.*)
28. Write a program to show how increment works with pointers. (*Hint: Use array and pointer arithmetic.*)
29. Write a program to compare:

```
*p++
(*p)++
```

(*Hint: Pointer increment vs value increment.*)

30. Write a program to continuously decrement user balance until it becomes insufficient. (*Hint: Use loop with subtraction and `--`.*)
31. Predict the output:

```
int i = 1;
printf("%d", i++++);
```

(Hint: Syntax error concept.)

32. Write a program that increments digits of a number one by one. (Hint: Extract digits using % and /.)
33. Write a program to generate natural numbers using post-increment. (Hint: Print then increment.)
34. Write a program to generate squares of numbers while incrementing loop variable. (Hint: Use $i*i$.)
35. Write a program to decrement temperature until freezing point is reached. (Hint: Use condition-based loop.)
36. Write a program to increment matrix elements row-wise. (Hint: Use nested loops.)
37. Write a program to decrement matrix diagonal elements only. (Hint: Condition $i == j$.)
38. Write a program to count vowels in a string using pointer increment. (Hint: Move pointer with $ptr++$.)
39. Write a program to reverse a string using decrement operators. (Hint: Start from end index.)
40. Write a program to demonstrate increment in for loop variations. (Hint: Try $i++$, $++i$, and $i+=1$.)
41. Write a program to decrement stock quantity after every sale. (Hint: Reduce count in loop.)
42. Write a program to increment marks of students by grace marks. (Hint: Array processing.)
43. Write a program to find how many increments are needed to reach a target number. (Hint: Counter variable.)
44. Write a program to decrement fuel level after every kilometer traveled. (Hint: Simulation logic.)
45. Predict the output:

```
int a = 5;  
int b = --a + a--;
```

(Hint: Trace step by step.)

46. Write a program to print Fibonacci series using incrementing loop variables. (Hint: Focus on loop increment.)
47. Write a program that increments only prime numbers in an array. (Hint: Prime checking function.)
48. Write a program that decrements salary until tax condition is satisfied. (Hint: Use loop with condition.)
49. Write a program to simulate elevator floors increasing and decreasing. (Hint: Use menu input.)
50. Write a program to understand operator precedence involving increment/decrement. (Hint: Mix arithmetic and increment operators in expressions.)